Djordje Batic

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AI Research Engineer with 5+ years of experience dedicated to accelerating scientific discovery through novel Deep Learning models. Proven track record in developing and applying techniques from Graph Neural Networks, Generative Models, and Explainable AI to model complex physical systems. Published author in top-tier journals and conferences (IEEE, Elsevier, ICASSP), with end-to-end experience leading research from ideation to large-scale deployment and publication.

Education

- **Ph.D. in Electronic & Electrical Engineering - University of Strathclyde** — Glasgow, UK
  ML for scientific applications at the intersection of AI and Energy Research | Marie Curie Fellow — *Oct 2021 - May 2025*
- **M.Sc. in Electrical & Computer Engineering - University of Novi Sad** — Novi Sad, Serbia
  Focus: Computer Science, ML, Robotics, Control Systems, Automation | CGPA: 9.50/10.00 — *Sep 2016 - Sep 2021*

Work Experience

- **Lloyds Banking Group** — Edinburgh, UK
  Senior Data & AI Scientist — *Jan 2026 – Present*
  - Working on AI Safety, focusing on AI risk, governance, and guardrails for generative and agentic AI use cases.
- **University of Strathclyde** — Glasgow, UK
  Marie Curie Early Stage Researcher — *October 2021 – Aug 2025*
  - **Explainable Knowledge Distillation for Edge AI / System Benchmarking and Observability:**
    - Achieved up to 22.6% predictive performance improvement by proposing a novel knowledge distillation algorithm that jointly improves robustness and interpretability of models used in edge deployment of energy disaggregation systems, while reducing the model size up to 76% [1,4].
    - Proposed and built a novel model-agnostic training framework and comprehensive benchmark suite for quantitative assessment, enhanced training and evaluation of AI system trustworthiness, focusing on measuring and improving robustness, faithfulness, and complexity metrics [2,3].
    - Developed a novel training framework that utilizes attribution priors to learn inherently explainable models, jointly enhancing interpretability and predictive performance in energy disaggregation [4].
  - **Applied ML & Research:**
    - Applied GNNs to the problem of optimal EV charging infrastructure placement, combining graph embeddings with a model-agnostic greedy search algorithm for efficient and scalable planning, achieving up to 55% improvement in charging utilization simulations. | Github [5–7]
    - Created and open-sourced GridCharge dataset, capturing over 5 million public EV charging sessions in Scotland with high-granularity data including carbon intensity, occupancy, energy consumption, and weather [7].
    - Leveraged LLMs to automatically parse, structure, and standardize thousands of diverse and unstructured tariff data documents. | Github | HuggingFace [7]
  - **Consultancy:**
    - **PictureIT**: Led a team developing a speech-to-image framework for an educational app that enables generation of custom AAC-style images for use in education of neurodivergent children. Implemented by fine-tuning a lightweight adapter over a diffusion-based generative model for multi-modal image generation.
    - **Hugo Energy**: Developed an automated pipeline for energy profiling and identification of electricity consumption flexibility using historical smart meter data collected from 50,000+ customers across UK.
    - **Flock Energy**: Led a team developing a core pipeline for training of energy disaggregation algorithms with contextual data used to deliver detailed energy usage insights. | Github
- **BioSense Institute** — Novi Sad, Serbia
  Research Assistant — *Jan 2021 – Oct 2021*
  - Developed a deep learning model using a multi-task learning approach for joint boundary delineation and instance segmentation of agricultural parcels from satellite imagery.
  - Achieved state-of-the-art performance on crop monitoring compared to single-task baselines.
  - **Master Thesis:** *"Multi-Task Learning for Joint Boundary Delineation and Segmentation of Agricultural Parcels"*
- **Cinteraction**
  AI Development Consultant — *Aug 2020 – Nov 2020*
  - Achieved 57% improvement in throughput on edge devices by training and optimizing inference of a CNN model for real-time facial affect analysis, leveraging CUDA inference optimization using double-buffering [10].
  - **Bachelor Thesis:** *"Facial Affect Analysis on Mobile Devices Using Convolutional Neural Networks"*. | Github

Skills

- Python, Pytorch, TensorFlow, C/C++, Cuda, Java, SQL, Git, Docker, Slurm, AWS, LaTeX

Awards and Funding

- **2021:** Marie Curie Doctoral Fellowship (∼€125,000) - EU Horizon 2020 MSCA-ITN Program
- **2023-2025:** Secured over £30k from consultancy projects, contributed to additional £200k funding from proposals.

Languages

- English (Fluent), Serbo-Croatian (Native), German (Basic)

References

[1] **Batic, D.**, Tanoni, G., Stankovic, L., Stankovic, V., Principi, E., "Improving Knowledge Distillation for Non-Intrusive Load Monitoring through Explainability Guided Learning", *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023.

[2] **Batic, D.**, Stankovic, V., Stankovic, L., "XNILMBoost: Explainability-informed Load Disaggregation Training Enhancement using Attribution Priors", *Engineering Applications of Artificial Intelligence*, 2024.

[3] **Batic, D.**, Stankovic, V., Stankovic, L., "Towards Trustworthy Load Disaggregation – A Framework for Quantitative Evaluation of Explainability using Explainable AI", *IEEE Transactions on Consumer Electronics*, 2023.

[4] **Batic, D.**, Tanoni, G., Principi, E., Stankovic, L., Stankovic, V., Squartini, S., "Interpretability and Reliability-driven Knowledge Distillation for Non-intrusive Load Monitoring on the Edge", *Expert Systems with Applications*, 2025.

[5] **Batic, D.**, Stankovic, V., Stankovic, L., "Geodemographic Aware Electric Vehicle Charging Location Planning for Equitable Placement Using Graph Neural Networks: Case Study of Scotland Metropolitan Areas", *Energy*, 2025.

[6] **Batic, D.**, Stankovic, V., Stankovic, L., "ChargeDEM: Geodemographic Aware EV Charging Infrastructure Placement for Enhanced Site Selection using Graph Neural Networks", *International Conference on Energy Efficiency in Domestic Appliances and Lighting (EEDAL)*, 2024.

[7] **Batic, D.**,"GridCharge: Capturing Carbon Intensity of 5M+ Public EV Charging Sessions in Scotland", 2025.

[8] **Batic, D.**, Stankovic, L., Stankovic, V., "Analysis of Smart Meter Data for Energy Waste Management", *Artificial Intelligence for Sustainability: Innovations in Business and Financial Services*, 2024.

[9] **Batic, D.**, "Towards trustworthy AI systems for Smart Grid management : facilitating robustness, transparency and fairness in the energy transition", *PhD Thesis*, University of Strathclyde, 2025.

[10] Ilic, V., **Batic, D.**, Mirkovic, M., Vukmirovic S., Culibrk, D., Bosakov, G., "Automatic Emotion Detection as a Teaching Aid in Online Knowledge Assessment", *2021 20th International Symposium INFOTEH*, 2021.

[11] **Batic, D.**, Culibrk, D., "Identifying Individual Dogs in Social Media Images", *2019 British Machine Vision Conference (BMVC) Workshop on Visual AI and Entrepreneurship (VAIE)*, 2019.